

Lab Exercise 19- ClusterIP in Service in Kubernetes

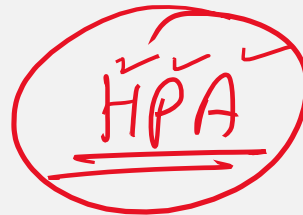
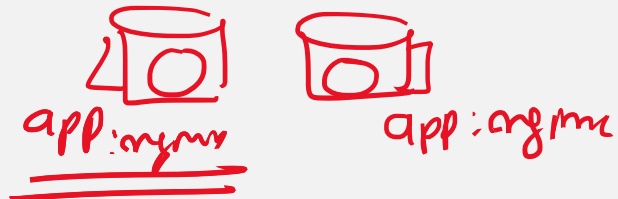
A ClusterIP is the default type of service in Kubernetes. It provides a stable internal IP address for the service, which is accessible only within the cluster. This is useful for internal communication between services within the cluster.

Step 1: Create Deployment (Correct Labels)

Create file:

nginx-deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx # MUST match template labels
  template:
    metadata:
      labels:
        app: nginx # MUST match service selector
    spec:
      containers:
        - name: nginx
          image: nginx
          ports:
            - containerPort: 80
```



Save file.

```
kubectl apply -f nginx-deployment.yaml
```

Verify:

```
kubectl get pods --show-labels
```

You should see:

```
app=nginx
```

Step 2: Create ClusterIP Service (Matching Selector)

Create file:

nginx-service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  type: ClusterIP
  selector:
    app: nginx # MUST match pod label
  ports:
    - port: 80
      targetPort: 80
```

Apply:

```
kubectl apply -f nginx-service.yaml
```

Step 3: Verify It Is Working

Check Service

```
kubectl get svc
```

You should see:

```
TYPE      CLUSTER-IP
ClusterIP  10.x.x.x
```

MOST IMPORTANT CHECK

```
kubectl get endpoints nginx-service
```

Now you should see something like:

```
10.244.x.x:80, 10.244.x.x:80
```

Step 4: Test From Inside Cluster

Run test pod:

```
kubectl run test-pod --image=busybox --rm -it -- sh
```

Inside pod:

```
wget -qO- http://nginx-service
```

You should see:

```
Welcome to nginx!
```

Exit:

```
exit
```

Step 5: Test From Outside (It Will Fail)

From your local machine:

```
curl http://<CLUSTER-IP>
```

It will NOT work.

That proves:

ClusterIP works only inside cluster.

```
Microsoft Windows [Version 10.0.26280.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NW>minikube status
minikube
type: Control Plane
host: Running
kublet: Running
apiserver: Running
kubeconfig: Configured

C:\Users\NW>(echo apiVersion: apps/v1 & echo kind: Deployment & echo metadata: & echo   name: nginx-app & echo spec: & echo   replicas: 2 & echo   selector:
& echo   matchLabels: & echo     app: nginx & echo   template: & echo     metadata: & echo       labels: & echo         app: nginx & echo     spec: & echo       e
cho     containers: & echo       - name: nginx & echo         image: nginx & echo         ports: & echo           - containerPort: 80 > nginx-deployment.ya
ml

C:\Users\NW>kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-app unchanged

C:\Users\NW>kubectl get pods --show-labels
NAME                                READY  STATUS   RESTARTS   AGE   LABELS
my-pod                             1/1    Running   2 (6m50s ago)  4h14m  app=web
nginx-app-75fdcbbc74-7jtf8         1/1    Running   0           82s   app=nginx,pod-template-hash=75fdcbbc74
nginx-app-75fdcbbc74-pvr42         1/1    Running   0           82s   app=nginx,pod-template-hash=75fdcbbc74
pod-with-secret                    0/1    CreateContainerConfigError 0       10m   <none>
test-pod                           1/1    Running   0           54s   run=test-pod

C:\Users\NW>(echo apiVersion: v1 & echo kind: Service & echo metadata: & echo   name: nginx-service & echo spec: & echo   type: ClusterIP & echo   selector:
& echo   app: nginx & echo   ports: & echo     - port: 80 & echo     targetPort: 80 > nginx-service.yaml

C:\Users\NW>kubectl apply -f nginx-service.yaml
service/nginx-service unchanged

C:\Users\NW>kubectl get svc
NAME      TYPE      CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
kubernetes  ClusterIP  10.96.0.1     <none>       443/TCP    4h14m
nginx-service ClusterIP  10.110.56.251 <none>       80/TCP     82s

C:\Users\NW>kubectl get endpoints nginx-service
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
```

```

C:\Users\NK>wget -qO- http://nginx-service
'wget' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\NK>kubectl delete pod test-pod
pod "test-pod" deleted from default namespace

C:\Users\NK>kubectl run test-pod --image=busybox --rm -it -- sh
All commands and output from this session will be recorded in container logs, including credentials and sensitive information passed through the command prompt.
If you don't see a command prompt, try pressing enter.
/ # kubectl run test-pod --image=busybox --rm -it -- sh
sh: kubectl: not found
/ # / #
sh: /: Permission denied
/ # wget -qO- http://nginx-service
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

```

```

font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
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For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
/ # exit
Session ended, resume using 'kubectl attach test-pod -c test-pod -i -t' command when the pod is running
pod "test-pod" deleted from default namespace

C:\Users\NK>nginx-service 10.244.0.71:80,10.244.0.72:80
'nginx-service' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\NK>|

```